

Code Layout, Readability & Reusability

Project Title: Rising Waters

Introduction

The Rising Waters project is developed by following good coding standards to make the application simple, readable, and easy to maintain. The source code is organized into separate files for machine learning, backend development, frontend design, and static resources.

Code Layout

The project follows a structured folder organization. Templates contain HTML pages, the static folder stores CSS and JavaScript files, the models folder stores the trained Machine Learning model and scaler, and the main application logic is implemented in the Flask application file. This organization improves code management and makes future updates easier.

Readability

The code uses meaningful variable names, proper indentation, comments, and consistent formatting. Each function performs a specific task, making the code easier to understand and debug. Error handling and logging are also included to improve reliability.

Reusability

Reusable functions are created for loading the model, preprocessing input data, and making predictions. These functions can be used in different parts of the application without rewriting the same code. This reduces duplication and improves maintainability.

Advantages

- Well-organized project structure.
- Easy to read and understand.
- Reusable functions reduce code duplication.
- Simple to maintain and update.
- Supports future enhancements.

Conclusion

A well-structured code layout with readable and reusable components improves software quality. It makes the Rising Waters project easier to maintain, test, and extend with additional features in the future.